

Jason Kiheon Lim

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EDUCATION

Hong Kong University of Science and Technology (HKUST)

Sep 2022 – Jun 2028

B.Eng. in Electronic Engineering · Hong Kong · GPA 3.42 / 4.30

- Honors: Dean's List (2025–26 Fall) · Full Admission Scholarship (2022 – present)
- Relevant coursework: Computer Organization & Design · Programming with C++ · Signals & Communications · Electronic Circuits · Matrix Algebra · Multivariable Calculus

RESEARCH EXPERIENCE

HKUST UROP1100 — Hardware-Efficient LUT Neural Networks on FPGA

Feb 2026 – Present

Undergraduate Researcher · Advisor: Prof. Wei Zhang · Hong Kong

- Analyzed a PyTorch LUT-CNN training/pruning pipeline (similarity-based pairing, BIB regularization, sensitivity pruning) and fixed a threshold-tie pruning bug restoring exact sparsity control (682 → 691 of target).
- Ran 40+ training runs across sparsity / packing configurations; characterized the accuracy–hardware frontier (Spearman $\rho = -0.96$), reaching up to ~86% pin / ~81% slice reduction.
- Ported the workflow to RadioML 1-D IQ classification with bitplane preprocessing and custom LUTConv1d operators; a pruning-first staged schedule held 40% vs 29% accuracy through full hardening.
- Built failure-analysis tooling (class/SNR breakdowns, confusion, confidence-margin) diagnosing structured error modes.

HKUST UROP1000 — Open-Source Digital IC Design Flow

Jun 2023 – Sep 2023

Undergraduate Researcher · Hong Kong

- Designed Verilog FSM architectures (clocked adder, finite-state-machine digital lock) for an open-source digital IC project.
- Took an RTL-to-GDS design through synthesis and place-and-route on the Tiny Tapeout platform.

PROJECTS

Systolic Array Accelerator

Verilog + Python

4x4 array with cycle-level trace visualization; measured 13 vs 12-cycle control overhead.

LUT vs MAC Inference Pipeline (RTL)

Verilog

Controlled datapath comparison with event-based latency; LUT 4 / MAC 5 cycles across all 18 runs.

Constraint-Aware Hardware Design Tool

Deployed web app

Constraint-first decision system: feasibility filter → Pareto reduction → explainable recommendation (Streamlit).

EXTRACURRICULAR & CERTIFICATIONS

GE Aerospace Job Simulation (Forage)

Jan 2026

Developed verification procedures for system reliability under environmental stress; built a signal-flow-based troubleshooting framework.

HKUST Korean Student Association Basketball Team

Sep 2022 – Present

Vice-Captain — led practices and coordinated logistics for the SKETCH HK Korean Basketball League.

POSTECH Science & Engineering Ideathon

Jun 2022

Participant & Awardee — "Most Creative and Extensive Idea" for an AI infant-monitoring concept.

MILITARY SERVICE

Republic of Korea Army — Capital Defense Command

Jan 2024 – Jul 2025

Supply Specialist, 1st Security Battalion · Seoul, Korea

- Managed logistics and equipment supply for a battalion defending core areas of the capital; coordinated inventory tracking and distribution to maintain operational readiness.

SKILLS

LANGUAGES — English (native) · Korean (native) · Mandarin (intermediate)

TECHNICAL — C++ · Python · Verilog HDL · PyTorch · GTKWave

SYSTEMS & HARDWARE — FPGA · pipeline design · latency analysis · model pruning

TOOLS — Linux · Git · Streamlit · Plotly · NumPy